

Recommended Design to Prevent Bending When Retrieving a Hand-Held Broom for Engineering Science Students

ESC101 Praxis

Word Count: 2915

1. Introduction	3
1.1 Understanding HHB Mechanisms	3
1.2 Opportunity Description	4
2. Tree Diagram	4
3. Requirements and Evaluation Criteria	4
4. Diagrams and Key Characteristics of Designs Considered	7
4.1 Suction Cup	7
4.2 Tripod Design	8
4.3 Weighted Base Design	9
4.4 Magnetic Stand Design	10
5. Comparison and Testing of Potential Designs	11
5.1 Calculation of Maximum Horizontal Force Resisted	11
5.2 Proxy Testing - Number of Steps to Retrieve Broom	12
5.3 Measurement Matrix	12
5.4 Pugh Charts for Evaluation Against Evaluation Criteria	13
5.5 Pairwise Comparison for Final Convergence	14
6. Conclusion	14
7. References	15
8. Appendices	18
Appendix A: Design Brief	18
Appendix B: Stakeholders Expectations	26
Appendix C: Interview Transcript	27
Appendix D: Requirements and Evaluation Criteria Justification & Calculations	28
Appendix E: Design Concept Key Decisions & Justifications	29
E.1: Three Key Tripod Design Decisions:	29
E.2: Four Key Weighted Base Design Decisions:	29
E.3: Five Key Magnetic Stand Design Decisions:	29
Appendix F: Horizontal Force Calculations	31
F.1: Suction Cup Calculations:	31
F.2: Tripod Calculations:	35
F.3: Weighted Base Calculations:	36
F.4: Magnetic Stand Calculations:	36
Appendix G: Proxy Test Design Descriptions	37
Appendix H: Pugh Charts	38
Appendix I: Pugh Chart Observations	41
Appendix J: Source Extracts	42

1. Introduction

This design report recommends an effective solution to eliminate the need for Engineering Science students to bend down to pick up the hand-held broom (HHB) when it falls. When individuals rest the HHB against a wall to temporarily set it aside, it may fall due to slight disturbances, requiring them to bend to retrieve it before continuing their cleaning. This process can be inconvenient and involves multiple steps. To address this splartz, this design report outlines the key features of four viable designs and compares them against a set of requirements and evaluation criteria informed by testing and research to determine the optimal design.

1.1 Understanding HHB Mechanisms

While the HHB is effective for cleaning, it remains unstable when left standing upright against a wall for convenience between uses. Because the broom has a small base, uneven bristles and a long slender handle, the broom is prone to tipping with slight disturbances (see Figure 1). Once it falls, the user must bend down to pick it up to continue using the broom. Picking up the broom requires five steps: bend over, grasp broom, lift broom, stand up and rotate broom (refer to section 5.2 for justification on these steps).



Figure 1. HHB falling after being rested against a wall

1.2 Opportunity Description

Bending down to retrieve a fallen broom off the wall can be annoying to some Engsci students. This presents an opportunity to design a broom that eliminates the need to bend when picking it up. The solution should prioritize remaining upright when placed anywhere and reducing the number of steps to retrieve the broom [see Appendix B for Stakeholder's Expectations].

2. Tree Diagram

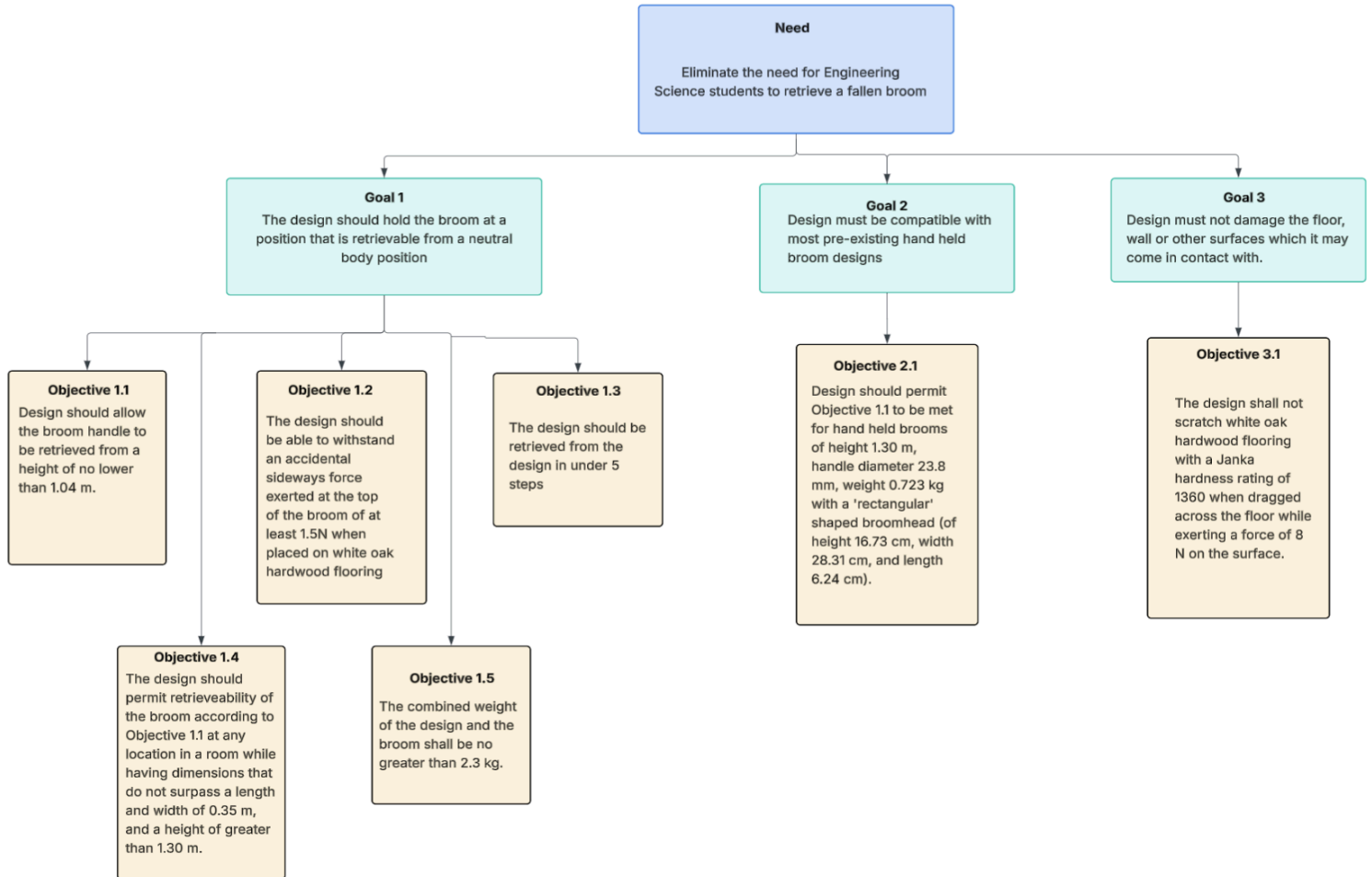


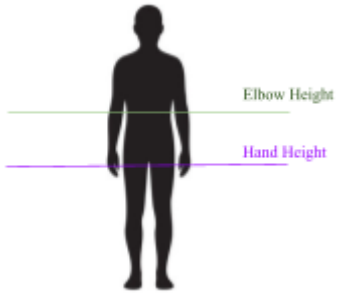
Figure 2. Tree Diagram of Need, Goals, and Objectives

3. Requirements and Evaluation Criteria

The following requirements and evaluation criteria were derived from the identified NGOs (Figure 2).

Table 1. Requirements and Evaluation Criteria

Goal 1: The design should hold the broom at a position such that the broom is retrievable from a neutral body position (NBP).
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<i>Requirement</i>	<i>Metric</i>	<i>Target</i>	<i>Evaluation Criteria</i>	<i>Justification</i>
R1.1 The design should allow the broom handle to be retrieved from a height of no lower than 1.04 m.	m	1.12		 Figure 3: Diagram depicting the relative height of the elbow and hand of an individual in an NBP [4]. The mean standing elbow height was found to be 1.122 m for men and 1.040 m for women [2]. Objects positioned at or above this height can be retrieved without bending, as they remain accessible through arm movement alone when the user is in a neutral body posture (Figure 3).
R1.2 The design should be able to withstand an accidental sideways force exerted at the top of the broom of at least 1.5 N when placed on white oak hardwood flooring.	N	1.5	EC1.2: A better design is one which allows the broom to withstand a greater accidental sideways force.	A 1.5 N minimum force was chosen because static-equilibrium analyses show that a broom leaning against a wall at a 75° angle will slip and fall under this load [Appendix D] [18][19]. 1.5 N is the baseline for accidental bumps resulting in a fallen broom, and any design must exceed it to perform better than simply leaning the broom against a wall.
R1.3 The broom should be retrieved from the design in under 5 steps	Number of steps	5	EC1.3: A better design is one which allows the broom to be released in as few steps as possible.	Using a proxy test, it took 5 steps to retrieve a broom from the ground. A valid solution must release the broom in under 5 steps to be more convenient and perform better than picking up the broom from the ground. Refer to Section 5.2 for evidence and justification of this 5 step broom

				retrieval.
R1.4 The design should permit retrieveability of the broom according to R1.1 at any location in a room while having dimensions that do not surpass a length and width of 0.35 m and a height of greater than 1.30 m.	m	Length = 0.35m Width = 0.35m Height = 1.30m	EC1.4: A better design is one which permits retrieveability of the broom according to R1.1 at any location in a room while taking up the smallest possible total volume.	The average broom head width and height of the HHB reference designs considered were approximately 0.35 m and 1.30 m, respectively [12-16]. The design should preserve the compactness of the broom and avoid increasing the surface area it takes up on the floor at any point in a room.
R1.5 The combined weight of the design and the broom shall be no greater than 2.3 kg.	kg	2.3	EC1.5: A better design is one which permits the combined weight of the design and the broom to be less than 2.3 kg.	The ergonomics of the broom is improved if its overall weight does not exceed 2.3 kg given it will be used away from the body in a repetitive motion [14].

Goal 2: The design must be compatible with most pre-existing hand held broom designs.

<i>Requirement</i>	<i>Metric</i>	<i>Target</i>	<i>Evaluation Criteria</i>	<i>Justification</i>
R2.1 Design should permit Objective 1.1 to be met for hand held brooms of height 1.30 m, handle diameter 23.8 mm, weight 0.723kg with a 'rectangular' shaped broomhead (of height 16.73cm, width 28.31cm, and length 6.24cm [12-16].	m, mm, kg	Height = 1.30m Handle Diameter = 23.8mm Weight = 0.723kg	EC2.1: A better design is one which permits Objective 1.1 to be met for HHB dimensions of the following range: height 1.40 to 1.10 m, diameter 22.0 to 25.4 mm, weight 0.460 to 0.907 kg and rectangular shaped broomhead dimensions of the following range: height 12 to 24.13 cm, width 14 to 37cm, and length 5 to 8.89 cm [12-16].	These dimensions cover a wider range of height, handle diameter, and weight metrics of the HHB reference designs. This increases the likelihood that the proposed designs will be compatible with most existing HHBs [12-16].

Goal 3: The design must not damage the floor, wall, or other surfaces which it may come in contact with.

<i>Requirement</i>	<i>Metric</i>	<i>Target</i>	<i>Evaluation Criteria</i>	<i>Justification</i>
R3.1 The design shall not scratch white oak hardwood flooring with a Janka hardness rating of 1360 when dragged across the floor while exerting a force of 7.09 N on the surface [16-18].	N	7.09	EC3.1: A better design is one which requires a force of greater than 7.09 N before creating a scratch on the hardwood flooring.	The Janka hardness rating of white oak falls roughly in the middle of other common hardwoods used for flooring [16-18]. The average weight of the reference broom designs considered was approximately 0.723 kg, corresponding to a downwards gravitational force of 7.09 N [17].

4. Diagrams and Key Characteristics of Designs Considered

4.1 Suction Cup

This design consists of a moveable nitrile suction cup that allows the broom to be secured to the floor when not in use. When deployed using a foot-actuated handle, the suction cup is released overtop of the broom bristles such that enough rubber is protruding underneath to create a firm seal with white oak hardwood flooring (Figure 4), allowing the broom to be held upright.

Three Key Design Decisions:

- 1) A foot-actuated handle enables the deployment and retractability of the suction cup from a standing neutral position, eliminating annoyance or discomfort associated with bending. The handle is made of aluminum for its ideal characteristics [Appendix E.1] [20].
- 2) Nitrile was chosen as the optimal material due to its increased tensile strength and resistance to scratches. Its high physical durability will help lengthen the lifespan of the design and allow it to withstand regular wear [27]. Most nitrile compounds have a lower density compared to silicon [30], lowering the weight of the design and improving its performance against EC1.5. It also does not tend to scratch most surfaces, improving its performance against EC3.1.
- 3) The dimensions were chosen to satisfy R1.4, allowing the design to minimally extend outwards from the broomhead such that it does not compromise compactability (EC1.4). The width of the suction cup was hence chosen to be 35.00 cm, the maximum permissible width to ensure equitable testing against R1.2, allowing for the suction cup to extend outwards from the broomhead by approximately 3.34 cm (Figure 4). The following dimensions were calculated so as to keep this outwards extension roughly uniform along the entire perimeter of the broomhead, resulting in a length of 12.93 cm when viewed from the side (Figure 4) and a depth of 2.00 cm, causing the total height of the design to be 18.73cm.

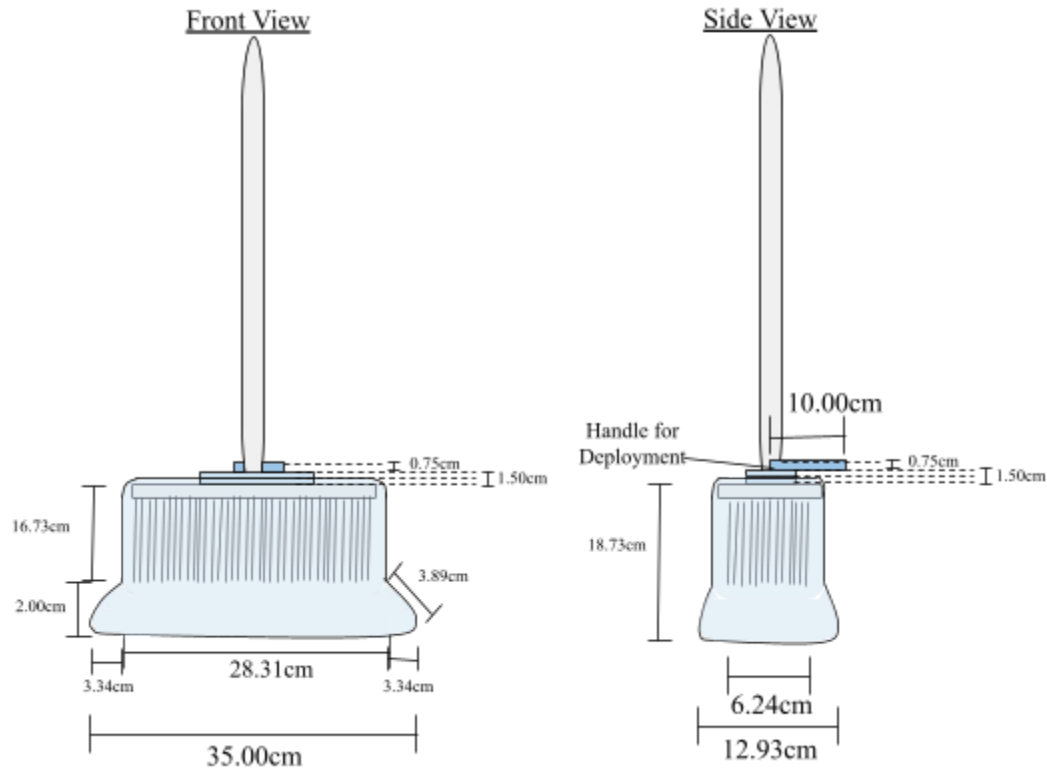


Figure 4. Dimensions of suction cup design while in its deployed position

4.2 Tripod Design

The tripod attachment is a three-legged, aluminum collapsible system in which all legs meet at an apex encircling the broom handle [see Appendix F.1 for tripod design decisions]. In the closed position, the legs fold inward so that their bottoms align with the bottom of the broom handle (see Figure 5). When the user presses the foot-actuated lever of length 0.10 m, the apex slides to the bottom of the broom handle and the three legs automatically deploy outward into a stable, open position. Each leg is 0.23 m long, set at a 40.6° deployment angle relative to the horizontal, with a leg height of 0.15 m, a diameter of 20 mm, and a maximum horizontal reach of 0.175 m from the center.

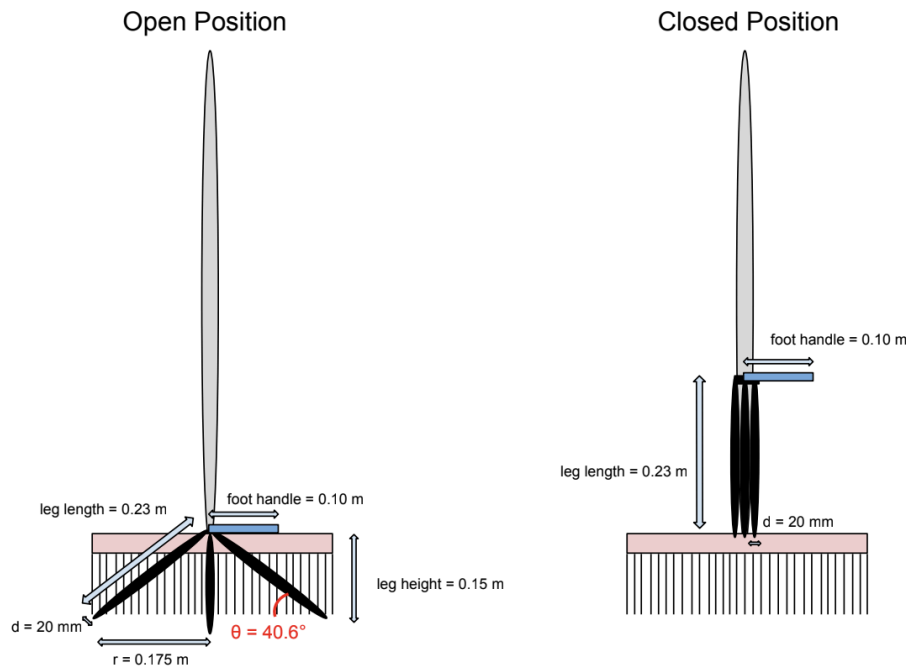


Figure 5. Tripod design and dimensions

4.3 Weighted Base Design

The weighted base design is a 35 cm x 35 cm x 35 cm block made of High-Density Polyethylene that weights 1.6 kg. A 25 cm x 2.38 cm x 2.38 cm central cutout allows the broom handle to be inserted (see Figure 6). The inner surface of this cutout, lined with silicon rubber, allows users to semi-permanently attach the device to the broom [see Appendix E.2 for weighted base design decisions]. To use the base, the user needs to flip the broom 180° so the base rests on the ground and flips it again to continue sweeping.

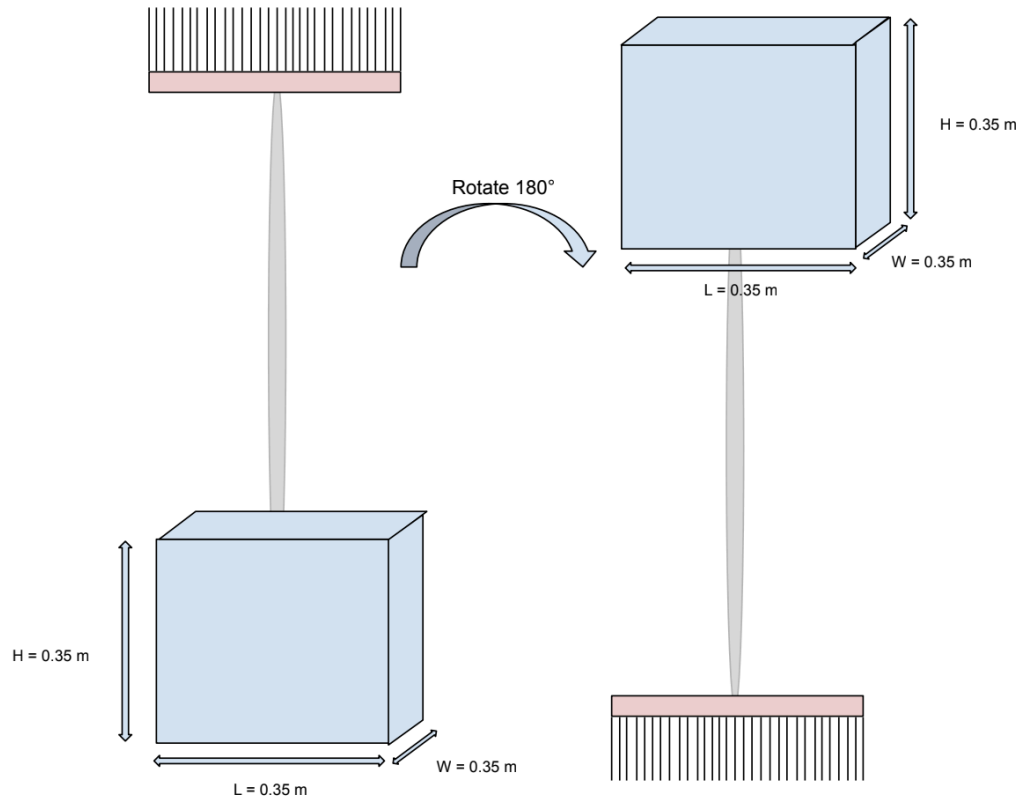


Figure 6. Weighted base design and dimensions

4.4 Magnetic Stand Design

The magnetic stand consists of a $35 \text{ cm} \times 35 \text{ cm} \times 35 \text{ cm}$ base with a $95 \text{ cm} \times 2.5 \text{ cm}$ pole attached to the top (see Figure 7). The stand is made of High-Density Polyethylene and weighs 1.04 kg. A Neodymium magnet, 2.5 cm in diameter and 5 mm thick, is attached to the top of the pole, and another magnet can be attached to the broom using Gorilla Glue [see Appendix E.3 for magnetic stand design decisions]. To use the stand, the user simply brings the magnetic side of the broom into contact with the magnet on the pole.

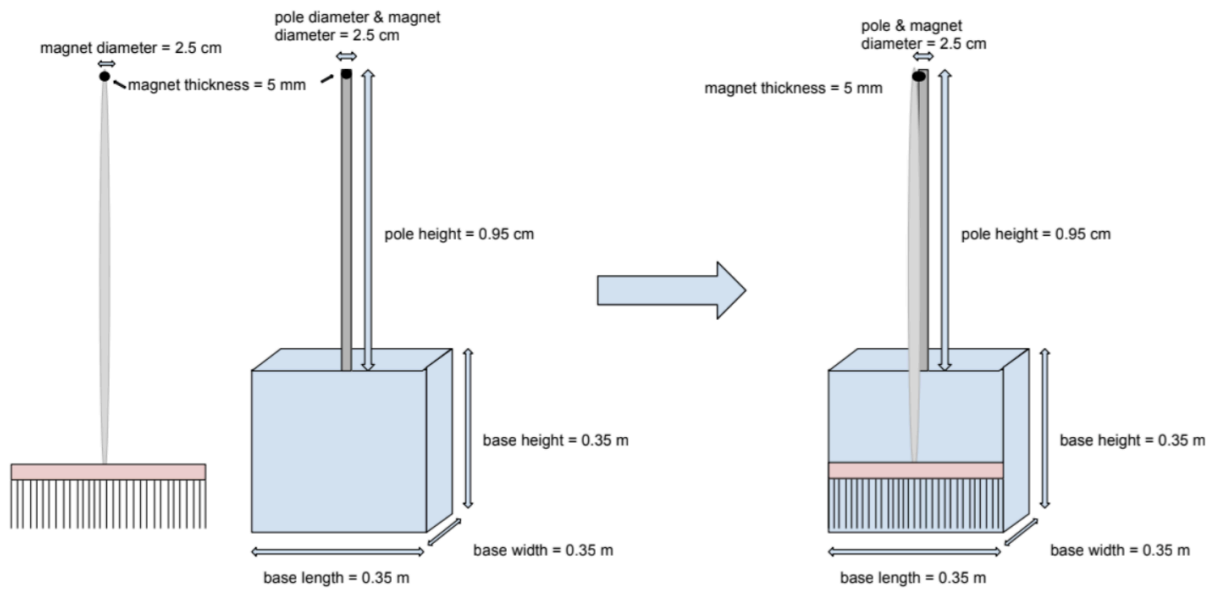


Figure 7: Magnetic stand design and dimensions

5. Comparison and Testing of Potential Designs

The most critical metric is the maximum horizontal force the broom design can withstand (R1.2). The maximum horizontal force is critical because it represents the amount of horizontal force the broom can resist before falling, which would require the user to bend down, failing to address the splartz. The second most critical metric is the number of steps required to retrieve the broom (R1.3). The number of steps to retrieve the broom is a critical metric because the tedious process of picking up a fallen broom makes it inconvenient.

5.1 Calculation of Maximum Horizontal Force Resisted

To permit equitable testing against R1.2, all design calculations assumed a base with dimensions not exceeding $0.35 \text{ m} \times 0.35 \text{ m}$ [9-13]. This enabled testing of the relative stability of the various design mechanisms themselves rather than having some designs demonstrate increased stability than others simply due to chosen base dimensions.

Refer to Appendix F for the full calculations of the horizontal force of all designs.

Suction Cup: The net downwards force on the broom consisted of the holding force of the suction cup and the combined weight of the suction cup and broom. By taking the sum of the moments about two potential tipping points of the broom, the maximum applied force was found to be 14.33 N [Appendix F.1].

Tripod: The minimum horizontal force required to tip the broom was determined by setting the overturning torque from the applied force equal to the restoring torque due to gravity. Solving this equation yielded a tipping force of 1.717 N [Appendix F.2].

Weighted Base: Using an overturning safety factor of 1.5, the combined mass of the broom and attachment, the height of the broom's centre of mass and the base width are used to solve for the allowable horizontal force. This calculation shows that the design can resist a horizontal force of 2.02 N without overturning [Appendix F.3].

Magnetic Stand: The stand's weight and base dimensions are used to calculate the resisting moment about its tipping edge. Dividing this moment by the height of the pole gives the maximum allowable horizontal force of 1.373 N [Appendix F.4].

5.2 Proxy Testing - Number of Steps to Retrieve Broom

The purpose of this test is to estimate the number of steps a user must take to retrieve the broom from each design (R1.3, EC1.3). Using reference designs to enable verification by similarity [see Appendix G for reference design descriptions], the results of the tests are summarized in Table 2 and demonstrated in this video: https://youtube.com/shorts/8Io3tpYN_xQ?feature=share

Table 2. The Steps Required to Retrieve the Broom From Each of the Four Designs

	Step Number	1	2	3	4
Designs	Suction Cup	Grab broom	Unplunge	Lift	Retract
	Tripod	Grab broom	Retract		
	Weighted Base	Grab broom	Lift	Flip	
	Magnetic Stand	Grab broom	Retract		

5.3 Measurement Matrix

Table 3. Measurement Matrix

	Design			
Metric	<u>Suction Cup</u>	<u>Tripod</u>	<u>Weighted Base</u>	<u>Magnetic Stand</u>
<i>Horizontal Force Resisted (N) (R1.2, EC1.2)</i>	14.33N	1.717	2.02	1.373
<i>Necessary Retrieval Steps</i>	4	2	3	2

<i>(Steps) (R1.3, EC1.3)</i>				
<i>Weight (kg) (R1.5, EC1.5)</i>	0.471	0.6	1.6	1.04

It was concluded the almost instantaneous nature of the magnetic base design eliminated the need for an additional deployment step, outperforming alternative designs against this metric. Furthermore, the suction cup exhibited strong resistance to horizontal force while adding minimal weight to the broom.

5.4 Pugh Charts for Evaluation Against Evaluation Criteria

Next, a series of Pugh charts were utilized to enable closer comparison between designs against specific evaluation criteria. The Pugh chart taking the suction cup design as the reference design is featured in Table 4.

Table 4. Pugh Chart with Suction Cup as Reference Design

	Design			
Evaluation Criteria	<u>Suction Cup (Reference)</u>	<u>Tripod</u>	<u>Weighted Base</u>	<u>Magnetic Stand</u>
<i>EC1.2: Sideways force resisted.</i>	0	–	–	–
<i>EC1.3: Number of retrieval steps.</i>	0	+	+	+
<i>EC1.4: Compactability of design.</i>	0	+	–	–
<i>EC1.5: Weight of design.</i>	0	–	–	–
<i>EC2.1: Compatibility with different brooms.</i>	0	+	+	+
<i>EC3.1: Damage to flooring, walls, or other contact surfaces.</i>	0	+	+	+
Results	N/A	4 + 2 –	3 + 3 –	3 + 3 –

The suction cup design suffers from limited compatibility with different brooms as it relies on being well-fitted to a particular broomhead to ensure a strong seal can be created with the floor. While the

current dimensions should accommodate average broomhead dimensions, variations in these dimensions would considerably weaken its performance against EC1.2. Additionally, while the nitrile material of the suction cup avoids floor scratches, the suction force exerted may cause some deformation in weaker materials.

Additional Pugh Charts can be viewed in Appendix H and observations are summarized in Appendix I.

5.5 Pairwise Comparison for Final Convergence

Taking into consideration the critical metrics and the conclusions drawn from the Pugh charts and measurement matrix, the pairwise comparison permitted a comprehensive holistic comparison of the designs.

Table 5. Pairwise Comparison

	<i>Suction Cup</i>	<i>Tripod</i>	<i>Weighted Base</i>	<i>Magnetic Stand</i>
<i>Suction Cup</i>	—	1	1	1
<i>Tripod</i>	0	—	1	1
<i>Weighted Base</i>	0	0	—	0
<i>Magnetic Stand</i>	0	1	1	—

While the weighted base meets R1.4, it differs from the other designs due to its mediocre performance against the critical metrics. While the tripod performed well with respect to compactibility, compatibility, and overall weight, its very poor performance against critical metric EC1.2 compared to the alternative designs led to its elimination. Although the magnetic stand design performs the best on EC1.3 and EC2.1, it performs poorly against EC1.2, which is the most critical metric, leading to its elimination. The suction cup design exhibited exceptional performance against critical metric EC1.2 while remaining lightweight and compact. While demonstrating poorer performance with respect to compatibility and steps required for retrieval, its ability to withstand horizontal force greatly reduces the chance a user will need to bend over to retrieve the broom if it has fallen, hence providing the best solution to the need.

6. Conclusion

In conclusion, the splartz outlined in this design report occurs when an HHB falls as a result of leaning it against a wall, requiring the individual to bend to retrieve it before continuing their cleaning. The design report explores four key design concepts: tripod design, weighted base design, magnetic stand design, and suction cup design. These designs are compared against a set of requirements and evaluation criteria informed by testing and research. Using various converging tools including Pugh charts, a Measurement matrix, and a Pairwise comparison, the suction cup design performed the best and was selected as the optimal solution to the splartz.

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8. Appendices

Appendix A: Design Brief

Design Optimization of a Hand-Held Broom for Users with Mobility Restrictions

1. Introduction

This design brief outlines the limitations of the everyday hand-held broom (HHB), emphasizing the posed harm for individuals with mobility restrictions to pick up the broom when it has fallen. When individuals accidentally drop the broom or want to momentarily rest it to do another task, they would inevitably need to pick it up to continue their cleaning. The bending motion to pick up the broom from the floor can be uncomfortable for those with mobility issues, causing unnecessary stress to their limbs. This design brief provides a document of requirements to improve the current design of the HHB to inspire design solutions that prioritize accessibility and safety.

1.1 Understanding HHB Mechanisms

The HHB is a common household cleaning tool designed to sweep dust, dirt, and debris from floors. The current design typically consists of two main parts: a long handle made of wood, metal, or plastic, and a brush head made of bristles or synthetic fibers [1]. While the HHB is simple and effective for cleaning, it remains unstable when left standing upright. Brooms are typically rested against a wall or on the floor for convenience between uses. Since the broom has a low center of mass and uneven bristles, the broom is prone to tipping and falling with slight disturbances. Once it falls, the user must bend down to pick it up to continue using the broom.

1.2 Stakeholder Expectations

The primary stakeholders of the HHB are Engineering Science students with mobility restrictions. In Canada, nearly 2.7 million individuals aged 15 and older reported having a mobility disability in 2017 [2]. The HHB should be designed to accommodate users experiencing limited flexibility, reduced range of motion or discomfort when bending.

The current HHB design requires users to bend down to pick up the broom. For people with mobility restrictions, this movement can cause discomfort, pain, or potential injury due to limited range of motion and reduced stability [3]. Forward flexion, twisting and side bending have been directly linked to higher rates of back pain and spinal stress [4]. Ergonomic studies further reveal that awkward or unsupported postures can compress nerves and irritate tendons, while quick or forceful bending motions increase lower back pressure [5]. First-hand observation supports this issue. An Engineering Science student and varsity athlete at the University of Toronto reports experiencing consistent back soreness as a result of strenuous physical activity. They noted that after an active day, bending down to pick up a broom puts additional stress on their already sore back [Appendix A]. These findings support that the broom design should prioritize accessibility by eliminating the need to bend when picking it up.

The expectation of stakeholders is that despite modifications to the design of the HHB, safety will be maintained avoiding the addition of harmful chemicals or sharp edges. The broom may cause airborne dust particles that can irritate the respiratory system; however, levels are typically too low to cause significant harm [6].

1.3 Opportunity Description

For individuals with mobility limitations, bending down to retrieve a broom can cause mild to moderate lower back pain. This presents an opportunity to design a broom that eliminates the need to bend when picking it up. The solution should prioritize accessibility for individuals with mobility challenges and maintain safety allowing these users to complete this activity without pain or discomfort.

2. Identification of Needs, Goals, and Objectives

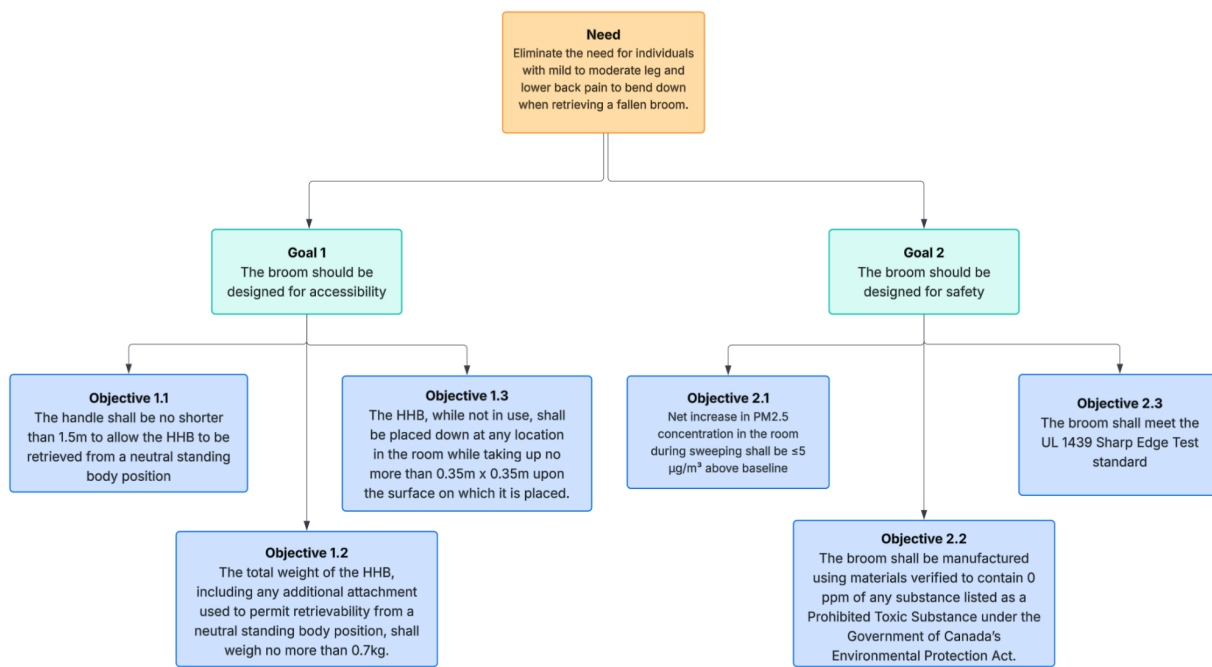


Figure 1. Visual depiction of the Needs, Goals, and Objectives for the modified HHB design.

Need

The design must eliminate the need for individuals with mild to moderate lower back pain to bend down when retrieving a fallen broom. The following goals and objectives outline the accessibility and safety considerations that must be implemented in the design solutions of the HHB.

Goals and Objectives

Table 1: A goal and four associated objectives for designing an accessible broom.

	Description	Metric	Target	Justification
Goal 1	An accessible design, for this brief, is one that allows individuals with mild to moderate lower back pain to use the broom in a manner that does not aggravate pre-existing lower back pain. This aligns with the definition for accessibility within the CSA-ISO9241-171:14, specifically to “achieve greater levels of [...] satisfaction that are as high as possible” in the context of sweeping with a broom (A2.1) [7].			
Objective 1.1 (O1.1)	The handle shall be no shorter than 1.5m to allow the HHB to be retrieved from a neutral body posture (NBP).	m	≥ 1.5	Following the definition of a NBP as defined in ISO11226:2000 (A2.2-3) [8], the standard asserts that “every effort should be made to obtain a working posture closer to the neutral posture,” (A2.2-3) [8,9]. The mean shoulder height for females and males in the United States of America was found to be 1.3425m and 1.4452m, respectively. (ISO/TR7250-2:2024, A2.4) [10]. Therefore, an HHB handle height of 1.5m will allow males and females (ISO/TR7250-2:2024, A2.4) [10] to seize the HHB while maintaining a NBP, hence limiting the aggravation of pre-existing back pain.
Objective 1.2 (O1.2)	The total weight of the HHB, including any additional attachment used to permit retrievability from a neutral standing body position, shall weigh no more than 0.7kg.	kg	≤ 0.7	Sweeping requires the repetitive motion of picking up and setting down the HHB using two hands, which can cause pain and fatigue. (ISO11228-3:2007, A2.5) [12] A mass of 0.7kg is consistent with alternative HHB designs (A2.8) [14-18] while far below the ISO standard to provide “comfort” when lifting for individuals aged 20-39 (ISO24533:2023, A2.6) [13].
Objective 1.3 (O1.3)	The HHB, while not in use, shall be placed down at any location in the room while taking up no more	m	≤ 0.35 m x 0.35m	The HHB user should be permitted to set the broom down on the floor anywhere provided there is sufficient space and retrieve it from a NBP. The average HHB width of alternative HHBs considered was 33.99cm

	than 0.35m x 0.35m upon the surface on which it is placed.			(A2.8) [14-18], hence this objective is consistent with existing alternatives.
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Table 2: A goal and three associated objectives for designing a safe broom.

	Description	Metric	Target	Justification
Goal 2	The broom should be designed for safety. In the context of this design brief, safety means that the broom is not likely to produce adverse health effects, which aligns with the international ISO/IEC Guide 51:2014 (A2.9) [19].			
Objective 2.1 (O2.1)	Minimize the net increase in fine particulate matter (PM2.5) concentration in the room.	mg/m ³	≤10	An increase ≤10 µg/m ³ above baseline was chosen because according to epidemiological exposure-response data, a 10 µg/m ³ increase in PM2.5 concentration is associated with a 0.65% rise in short-term health risk (A2.10) [20]. Therefore, this target is a conservative and health-protective metric for re-suspension control [AB2.1].
Objective 2.2 (O2.2)	Prohibit any listed toxic substance under the Government of Canada's Environmental Protection Act (A2.11) [21]	ppm	0	The limit of 0 ppm of toxic substances ensures compliance with CEPA and the Canadian Government's Prohibited Toxic Substances List [AB2.2]. Prohibiting the use of toxic substances ensures consumer safety by preventing exposure to endocrine-disrupting or carcinogenic compounds (A2.12) [22].

Objective 2.3 (O2.3)	The HBB shall meet the UL 1439 Sharp Edge Test standard.	The number of layers of UL sensing tape that get cut through when tested with a calibrated UL Sharp Edge Tester applying 6.67 N of force over a 50 mm stroke [AB2.3].	≤ 1 (pass test)	The UL 1439 Sharp Edge Test was selected because it provides a standardized, repeatable and Canadian industry-recognized method for evaluating the safety of product edges [AB2.3]. If a maximum of one layer of the UL sensing tape is cut, this confirms that all contact surfaces are sufficiently smooth and rounded to avoid cuts or abrasions during normal handling (A2.13) [23].
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3. Reference Designs

Reference Design 1: *Medium sized two piece broom and standing dust pan set (A3.1) [22]*

While the handle height of the broom meets O1.1, the accompanying dustpan falls 0.5m below the objective. The design is 0.44 kg heavier than in O1.2, but satisfies O1.3 when placed down in the arrangement shown. There is insufficient information to evaluate O2.1-3 (A3.1).

This design combines the HHB and dustpan, thereby improving the accessibility in two ways: the broom is supported upright by the dustpan's larger surface area, and debris can be collected without extensive bending. However, this design is heavier than most HHBs (A3.1), the dustpan handle is slightly shorter than ideal, and it requires purchasing a new set rather than integrating with existing HHBs.



Figure 2. Design of two piece broom and standing dust pan set

Reference Design 2: *Hands-free bicycle kickstand with rubber gripping (A3.2) [23]*

While not an existing HHB design, the kickstand has unique characteristics which pertain to O1.1-3. It enables the bike to remain upright and retrievable from a NBP. The kickstand adds minimal weight to the bicycle (0.3kg), does not protrude very far laterally (allowing it to be parked anywhere consistent with O1.3), and can be operated while maintaining an NBP (aligning with O1.1). Finally, it's designed to be easily added and removed from any bicycle allowing widespread compatibility.



Figure 3. Design of hands-free bicycle kickstand

Reference Design 3: *Broom clip holders that attach to the wall*
[24]

Self-installable HHB holder clips allow for customized installation heights so individuals can maintain a NBP while accessing an HHB, aligning with O1.1 but failing to meet O1.3. The design also requires permanent modification of the wall, which may be offputting for some users. The HHB holder clips support O1.2 by adding no extra weight to the HHB. Similar to the kickstand design, the clips provide compatibility with any existing HHB given it can be customized for different handle diameters.



Figure 4. Design of broom clip holder

4. Conclusion

In conclusion, there is an opportunity to eliminate the need for our primary stakeholders to bend down to pick up the broom. The solution to this opportunity must be a design that is accessible to individuals with mobility issues and be designed for safety, such that the broom is not likely to produce adverse health effects in users. By following the provided needs, goals, and objectives through the lens of our primary stakeholder, there is a broad avenue for a responding team to find a solution to our splartz.

Appendix B: Stakeholders Expectations

The primary stakeholders of the HHB are Engineering Science students who use brooms. An interview with an EngSci student and varsity athlete at the University of Toronto supports this claim [Appendix B]. They noted that when they have to clean after an active day, bending down to pick up a fallen broom is annoying and requires additional steps to pick it up. Stakeholders expect the broom to remain upright when placed anywhere, eliminating the need to bend down. They also expect that retrieving the broom does not require too many steps, as this is inconvenient and time-consuming.

Appendix C: Interview Transcript

Interviewer: Good afternoon. What is your name and what program are you in?

Interviewee: Hi, my name is [REDACTED] and I'm in UofT EngSci.

Interviewer: Very nice. Have you ever used a hand-held broom before?

Interviewee: Oh I use it pretty often.

Interviewer: Where do you typically use it to clean?

Interviewee: In my dorm or when I'm at home with my family.

Interviewer: When you're done with the broom, where do you typically put it?

Interviewee: I put it against the wall.

Interviewer: Have you ever experienced needing to grab the broom and finding it on the floor after sliding off the wall?

Interviewee: Oh my god, yes! That is actually the bane of my existence.

Interviewer: Haha. Why does this annoy you so much?

Interviewee: Well, I'm a varsity athlete here at UofT and I practice all the time. When I need to sweep to clean my room, I find myself needing to rest the broom so I can move things around and sweep there. And the easiest place to rest the broom is against the wall. But every time, without fail, I return to see the broom on the floor. And after being active for the entire day, I'm just so tired and the last thing I want to do is to have to constantly pick up a broom that accidentally falls on the floor.

Interviewer: Thank you so much for your time. We appreciate your input. We wish you a good day!

Interviewee: Yeah, no problem.

Appendix D: Requirements and Evaluation Criteria Justification & Calculations

Assumptions:

- 1) Assume the lean angle of broom to horizontal is 75° . This lean angle was chosen because it aligns with the widely recommended '4:1' ladder safety rule used by OSHA, HSE, and CCOHS, making it a realistic and evidence-based reference for how steeply household objects like brooms are typically leaned against walls [18].
- 2) Assume the static friction coefficient between the wood floor and plastic broom bristle is 0.4. This was chosen because reference data for common material pairings show that hard plastic on dry wood has a static coefficient of friction of roughly 0.4 [19].

Maximum sideways force the broom can withstand when leaning against a wall before it slips and falls:

$$\begin{aligned}F_{F,max} &= \mu N \\&= \mu mg \cos \theta \\&= (0.4)(0.7)(9.8) \cos 75^\circ \\&= 1.5 \text{ N}\end{aligned}$$

Appendix E: Design Concept Key Decisions & Justifications

E.1: Three Key Tripod Design Decisions:

- 1) The tripod legs were designed to extend a maximum of 0.175 m horizontally to meet R1.4, ensuring fair and consistent comparison with other concepts. Increasing the support base width (i.e., spreading tripod legs) increases resistance to toppling when horizontal force is applied [21]. Therefore, using the full allowable horizontal distance while still satisfying R1.4 maximizes the horizontal force the tripod can withstand before toppling, which is our most critical evaluation criteria.
- 2) When deployed, the apex of the tripod is at the bottom of the broom handle. This is advantageous, as a lower centre of gravity results in greater stability [21]. Therefore, this maximizes the amount of horizontal force the broom can withstand, which is our most critical evaluation criteria.
- 3) The material is aluminum because aluminum provides the optimal balance of strength, weight, and durability for a tripod intended for repeated loading and safe operation [20].

E.2: Four Key Weighted Base Design Decisions:

1. The dimensions of the weighted attachment were chosen to be 35 cm x 35 cm x 35 cm. This is the largest size that still meets Requirement R1.4 and allows for fair and consistent comparison with the other designs.
2. The weighted base will be made of High-Density Polyethylene (HDPE). This material is non-toxic, durable, and impact-resistant, making it safe and reliable for everyday use. It is also cost-effective, easy to recycle and maintains its strength in a wide range of temperatures, making it ideal for the base. [28]
3. According to the Canadian Centre for Occupational Health and Safety, a hand-held tool that is used away from the body or above shoulder height should not exceed 2.3 kg [29]. Because the weight of the broom is 0.7 kg, the maximum weight of the weighted base attachment can be 1.6 kg without causing the user pain.
4. The inner cutout will be lined with silicone rubber, chosen for its high friction, wide temperature resistance, and strong shock absorption. The high friction ensures the broom stays securely in place without requiring a permanent attachment to the base [30].

E.3: Five Key Magnetic Stand Design Decisions:

- 1) The stand has a mass of 1.04 kg, modeled after the SAVORSPACE dustpan due to its similar dimensions. Although the stand is not a dustpan, its design and mechanism are similar, making it familiar to users [32].
- 2) The bottom base of the stand will be 35 cm x 35 cm x 35 cm with a 95 cm x 2.5 cm pole attached to the top. The base is the largest size that meets Requirement R1.2 and the pole is the height of the broom and is the width of the broom.
- 3) The magnetic stand will be made of High-Density Polyethylene (HDPE). This material is non-toxic, durable, and impact-resistant, making it safe and reliable for everyday use. It is also cost-effective, easy to recycle and maintains its strength in a wide range of temperatures, making it ideal for the base. [28]
- 4)

- 5) For the magnetic broom stand, two Neodymium magnets will be used due to the high strength and compact size of Neodymium [33]. Two disc magnets, each measuring 2.5 cm in diameter and 5 mm in thickness will be used, as they can support a 0.7 kg broom. The magnets have a diameter of 2.5 cm to ensure that they can fit on the stand pole with a diameter of 2.5 cm. One magnet will be positioned at the top of the stand pole and the other on the top of the broom. These locations have been selected for usability, as they allow the user to attach the broom without bending, thereby satisfying Requirement R1.1.
- 6) Gorilla Glue (polyurethane) has been selected to attach the magnets to the stand and the broom due to its strength and durability. It can resist moisture, temperature changes and repeated stress, ensuring that the magnets remain secured during regular use. The glue is also safe for users to handle.

Appendix F: Horizontal Force Calculations

F.1: Suction Cup Calculations:

The design must accommodate broomheads of height 16.73cm, width 28.31cm, and length 6.24cm, which aligns with the average dimensions for the reference designs considered [Appendix G]. The width of the suction cup was hence chosen to be 35.00cm, the maximum permissible width to ensure equitable testing against R1.2, allowing for the suction cup to extend outwards from the broomhead by approximately 3.34cm. The following dimensions were calculated so as to keep this outwards extension roughly uniform along the entire perimeter of the broomhead. This required the length of the suction cup to be 12.93cm, as evident in Figure 1.

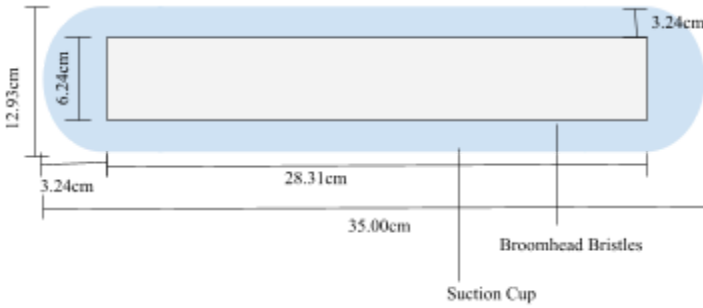


Figure 1. Bottom view of broomhead and suction cup dimensions

The area of the suction cup base was approximated as the combined area of two semicircular end pieces and the rectangular section in the middle.

$$\begin{aligned} A_{base} &= lw + \pi r^2 \\ A_{base} &= (28.31)(12.93) + \pi(12.93/2)^2 \\ A_{base} &= 366.0483 + 131.3067 \\ A_{base} &\approx 497.36 \text{ cm}^2 \end{aligned}$$

The holding force of a suction cup refers to the weight in newtons of the heaviest object the suction cup can be expected to hold. The holding force will change depending on the orientation and loading conditions of the suction cup [24], and so for this application the holding force of interest was that associated with lifting an object vertically upwards without horizontal motion. This was theoretically calculated by modelling the suction cup according to those manufactured by HPC Europe and following a similar calculation procedure to arrive at a plausible approximation [26]. A conservative approximation of a 30% vacuum level was chosen assuming the user doesn't exert excessive force when plunging the suction cup.

$$\begin{aligned} \text{Holding Force (N)} &= \text{Suction Cup Surface Area (cm}^2\text{)} \times \text{Vacuum level (\%)} \times 0.099366 \\ \text{Holding Force (N)} &= 497.36 \times 30 \times 0.099366 \\ \text{Holding Force (N)} &\approx 1482.62 \text{ N} \end{aligned}$$

Typically, a factor of safety (FOS) between 3 and 5 is applied to the maximum holding force [26], and so the average value of 4 was used.

$$\begin{aligned} \text{Holding Force}_{\text{with FOS}}(N) &= \text{Holding Force} / \text{FOS} \\ \text{Holding Force}_{\text{with FOS}}(N) &= 1\,482.62N / 4 \approx 370.66N \end{aligned}$$

If the suction cup can exert a maximum force of 14.83N upwards to lift an object, it can be assumed it pulls the broom downwards with the same force when deployed onto the floor.

Next, the approximate weight of the broom and suction cup were calculated to determine the total weight of the design and broom.

The design must accommodate brooms of mass 0.723kg, which aligns with the average weight metrics for the reference designs considered (Appendix F).

$$\begin{aligned} \text{Broom Weight}(N) &= m_{\text{broom}}g \\ \text{Broom Weight}(N) &= (0.723)(9.81) \approx 7.09N \end{aligned}$$

To determine the weight of the suction cup, the mass of nitrile used to create it was first determined.

Common materials used to make suction cups include synthetic rubbers such as nitrile and silicon. Comparing the two, nitrile generally demonstrates better resistance to scratches and tears. While the two can withstand similar compressive forces before they start to permanently deform, nitrile's tensile strength is about twice that of silicone, making it more resistant to stretching [27]. Therefore, nitrile was chosen to make the suction cup as its higher physical durability will help lengthen the lifespan of the design and allow it to withstand regular wear.

First, the surface area of the upper portion of the suction cup was calculated – that is, the portion which covers the broomhead bristles when in the deployed position. This was approximated to be the surface area of a rectangular prism with roughly the same dimensions as the broomhead as specified in Figures 1 and 2.

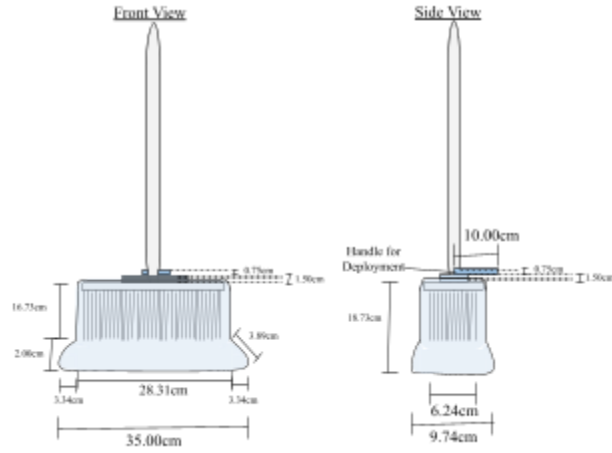


Figure 2. Dimensions of suction cup design while in its deployed position.

$$Surface Area_{Upper Suction Cup} (cm^2) = lw + 2hw + 2hl$$

$$Surface Area_{Upper Suction Cup} (cm^2) = (6.24)(28.31) + 2(16.73)(28.31) + 2(6.24)(16.73)$$

$$Surface Area_{Upper Suction Cup} (cm^2) \approx 1332.70cm^2$$

Next, the surface area of the lower portion of the suction cup was calculated – that is, the portion which extends outwards from the broomhead by approximately 3.34cm. This was approximated as a rectangular strip running around the perimeter of the broomhead. The width of the rectangular strip was taken as the perimeter of the broomhead. Following the proportions used by HPC Europe, to ensure the maximum holding force calculation method would still be applicable, the lower portion of the suction cup was determined to extend 2.00cm downwards from the broomhead bristles, allowing the remaining rectangle dimension to be calculated using the pythagorean theorem.

$$Perimeter_{Broomhead} (cm) = 2lw = 2(28.31)(6.24)$$

$$Perimeter_{Broomhead} (cm) \approx 353.31$$

$$Remaining Rectangle Dimension (cm) = \sqrt{3.34^2 + 2^2} \approx 3.89$$

$$Surface Area_{Lower Suction Cup} (cm^2) = lw = (353.31)(3.89) \approx 1374.38$$

The thickness of the rubber in the suction cup was chosen to be 0.2cm as this was conventional for the reference designs considered [25]. This allowed for the calculation of the total volume of rubber that would be needed to make the suction cup and its corresponding weight. The density of nitrile rubber ranges from 1.00 to 1.30g/cm³, and so the average value of 1.15g/cm³ was used for calculations [27].

$$Total Surface Area of Suction Cup (cm^2) = 1332.70 + 1374.38 \approx 2707.08$$

$$\begin{aligned} \text{Total Volume of Suction Cup (cm}^3\text{)} &= (2\,707.08)(0.2) \approx 541.42 \\ \text{Total Mass of Suction Cup} &= 541.42g \div \left(\frac{1.15g}{\text{cm}^3}\right) \approx 470.80g \text{ or } 0.471kg \end{aligned}$$

This allowed for the calculation of the total weight of the suction cup and broom.

$$\begin{aligned} \text{Total Mass of Suction Cup and Broom (kg)} &= 0.471 + 0.728 \approx 1.199kg \\ \text{Total Weight of Suction Cup and Broom (N)} &= (1.199)(9.81) \approx 11.76N \end{aligned}$$

This allowed for the calculation of the total downwards force being exerted on the broom.

$$\text{Net Downwards Force on Broom (N)} = \text{Total Weight of Suction Cup and Broom} + \text{Holding Force}_{\text{with FOS}}$$

$$\text{Net Downwards Force on Broom (N)} = 11.76 + 370.66 \approx 382.42$$

Knowing the total external forces acting on the broom allowed for the calculation of the force required to create a moment great enough to cause tipping of the broom about the furthest point along its width (Figure 3) and the furthest point along its length (Figure 4).

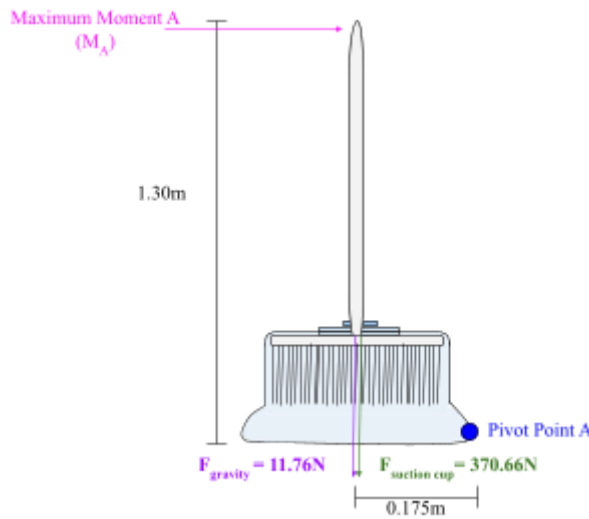


Figure 3. Free body diagram of broom and suction cup design when subjected to a moment at the furthest point along its width.

$$\begin{aligned} \Sigma M_A &= 0 \\ 0 &= -M_A(1.3) + (382.42)(0.175) \\ M_A &= 51.48N \end{aligned}$$

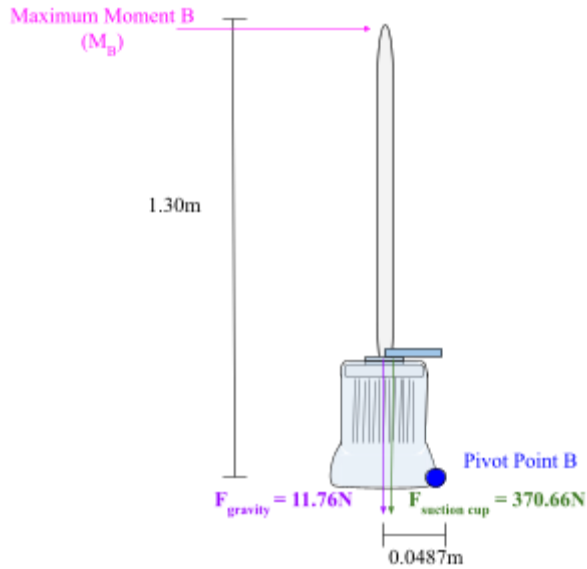


Figure 4. Free body diagram of broom and suction cup design when subjected to a moment at the furthest point along its length.

$$\begin{aligned}\Sigma M_B &= 0 \\ 0 &= -M_B(1.3) + (382.42)(0.0487) \\ M_B &= 14.33N\end{aligned}$$

F.2: Tripod Calculations:

Calculating Mass of Tripod

Solid cylinder volume:

$$\begin{aligned}V &= \pi r^2 L \\ &= \pi(0.01)^2(0.23) \\ &= 7.225 \times 10^{-5} m^3\end{aligned}$$

$$\begin{aligned}V_{total} &= 3V \\ &= 3(7.225 \times 10^{-5}) \\ &= 2.1675 \times 10^{-4} m^3\end{aligned}$$

$$\begin{aligned}m &= \rho V \\ &= (2700)(2.1675 \times 10^{-4}) \\ &= 0.6 kg\end{aligned}$$

Calculating Minimum Force for Tipping

Overturning torque: $\tau = F \times d$

Since the lever arm and the direction of the force are perpendicular: $\tau_{push} = Fh$

Restoring torque due to gravity: $\tau_{gravity} = F_g R$

Tipping occurs when: $Fh = F_g R$

Solve for F:

$$\begin{aligned} F &= \frac{mgR}{h} \\ &= \frac{(0.6+0.7)(9.8)(0.175)}{1.30} \\ &= 1.717 \text{ N} \end{aligned}$$

E.3: Weighted Base Calculations:

The overturning safety factor used in the calculations is 1.5. Although no standard safety factor exists specifically for brooms, this value adheres to typical code recommendations [31].

$$\begin{aligned} SF &= \frac{(m_{broom} + m_{attachment}) \times g \times b}{F_{lateral} \times h_{top}} \\ 1.5 &= \frac{(0.7 \text{ kg} + 1.6 \text{ kg}) \times 9.81 \times 0.175 \text{ m}}{F_{lateral} \times 1.3 \text{ m}} \\ F_{lateral} &= 2.02 \text{ N} \end{aligned}$$

E.4: Magnetic Stand Calculations:

It is assumed that the mass of the stand handle is negligible, as the centre of mass for the magnetic stand is halfway from the bottom of its base.

$$\begin{aligned} \tau_{resist} &= F_g \times \frac{b}{2} = (9.81)(1.04) \times \frac{(0.35)}{2} = 1.785 \text{ Nm} \\ \tau_{tip} &= F_{lateral} \times h \\ F_{lateral} &= \frac{\tau_{resist}}{h} = \frac{1.785 \text{ Nm}}{1.3 \text{ m}} = 1.373 \text{ N} \end{aligned}$$

Appendix G: Proxy Test Design Descriptions

Table 6. Reference Designs Used as a Proxy for the Four Designs

<u>Design</u>	<u>Reference Design Used as a Proxy</u>
<i>Suction Cup Attachment</i>	A plunger had similar characteristics to a suction cup, and allowed for reasonable approximation of its behaviour. While a foot-actuated handle was not available, steps relating to the use of this handle were accounted for separately.
<i>Tripod Attachment</i>	While a foot-actuated handle was not available, a traditional tripod was used to mimic the steps needed to deploy a similar tripod mechanism attached to a broom.
<i>Weighted Base</i>	The cylindrical handle of the plunger was attached to a flat tupperware to model a flat base. The remaining free end of the cylindrical handle symbolically represented the broomhead.
<i>Magnetic Stand</i>	Magnets were attached equally along the length of the broom handle. Magnets were affixed to a wall to mimic the broom's attachment to a sturdy vertical structure.

Appendix H: Pugh Charts

Table 7. Pugh Chart with Tripod as Reference Design

	Design			
Evaluation Criteria	<u>Suction Cup</u>	<u>Tripod (Reference)</u>	<u>Weighted Base</u>	<u>Magnetic Stand</u>
<i>EC1.2:</i> Sideways force resisted.	+	0	+	+
<i>EC1.3:</i> Number of retrieval steps.	–	0	0	+
<i>EC1.4:</i> Compactability of design.	–	0	–	–
<i>EC1.5:</i> Weight of design.	+	0	–	–
<i>EC2.1:</i> Compatibility with different brooms.	–	0	+	+
<i>EC3.1:</i> Damage to flooring, walls, or other contact surfaces.	–	0	–	–
Results	2 + 4 –	N/A	2 + 3 –	3 + 3 –

Table 8. Pugh Chart with Weighted Base as Reference Design

	Design			
Evaluation Criteria	<u>Suction Cup</u>	<u>Tripod</u>	<u>Weighted Base (Reference)</u>	<u>Magnetic Stand</u>
<i>EC1.2:</i> Sideways force resisted.	+	–	0	+
<i>EC1.3:</i> Number of retrieval steps.	–	0	0	+
<i>EC1.4:</i> Compactability of	+	+	0	–

design.				
<i>EC1.5:</i> Weight of design.	+	+	0	0
<i>EC2.1:</i> Compatibility with different brooms.	–	–	0	+
<i>EC3.1:</i> Damage to flooring, walls, or other contact surfaces.	–	+	0	0
Results	3 + 3 –	3 + 2 –	N/A	3 + 1 –

Table 9. Pugh Chart with Magnetic Stand as Reference Design

	Design			
Evaluation Criteria	<u>Suction Cup</u>	<u>Tripod</u>	<u>Weighted Base</u>	<u>Magnetic Stand (Reference)</u>
<i>EC1.2:</i> Sideways force resisted.	+	+	–	0
<i>EC1.3:</i> Number of retrieval steps.	–	–	–	0
<i>EC1.4:</i> Compactability of design.	+	+	+	0
<i>EC1.5:</i> Weight of design.	+	+	0	0
<i>EC2.1:</i> Compatibility with different brooms.	–	–	–	0
<i>EC3.1:</i> Damage to flooring, walls, or other contact surfaces.	–	0	0	0
Results	3 +	3 +	1 +	N/A

	3 –	2 –	3 –	
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Appendix I: Pugh Chart Observations

Table 10. Summary of Pugh Chart Observations

Base Design	Observations
Tripod (Appendix F)	• Cannot withstand much sideways force, but matches or outperforms other designs against most other evaluation criteria. • While performing slightly worse than the weighted base and magnetic stand against R2.1, its design depends highly on the diameter of the broom handle rather than the broomhead, which is less variable across different designs. • Relatively lightweight with only three small points of rubberized contact with flooring, allowing for excellent performance against EC3.1.
Magnetic Stand (Appendix F)	• Outperforms all designs on EC1.3 • Had the best performance in EC2.1 given the magnets could be implemented on any broom regardless of its dimensions entirely.
Weighted Base (Appendix F)	• Compatibility with various brooms is dependent on broom handle diameter rather than broomhead dimensions, enabling improved performance against EC2.1. This is its main differentiating factor from the magnetic stand, given they share similar base dimensions and weight and hence perform similarly against EC1.2 and 1.7.

Appendix J: Source Extracts

1.1 HHB Material Makeup [1]

Brooms have been used for centuries to sweep up, in, and around the home and workplace. They may be made from a variety of materials, both man-made and natural. Man-made bristles are generally of extruded plastic and metal handles. Natural-material brooms may be constructed of a variety of materials, including brush, but generally include stiff grasses such as broomcorn and/or sotol fiber.

Broomcorn brooms have been made for at least 200 years and are considered superior brooms. Plastic brooms merely move dirt around, however, broomcorn stalks actually absorb dirt and dust, wear extremely well, and are moisture-resistant. Broomcorn brooms are the most expensive of the manufactured brooms.

ISO 11226:2000 Definition of Neutral Body Posture and Recommendation to Maintain Working Posture Close to Neutral Posture Whenever Possible - Neutral posture for the trunk, upper arms, and head [2]

Upright trunk upper arms hanging freely, and head posture according to the Frankfurt plane.

Frankfurt plane [3]

Standard horizontal plane at the level of the upper edge of the opening of the external auditory meatus (external ear opening) and the lower border of the orbital margin (lower edge of the eye socket); the median plane of the head is held vertically.

NOTE the Frankfurt plane is associated with the normal line of sight (relaxed extraocular muscles).

Evaluation Procedure [4]

An “acceptable” evaluation result means that a working posture is acceptable only if VARIATIONS of posture are also present (see 3.1). In any eventuality, every effort should be made to obtain a working posture closer to the neutral posture, if this is not already the case.

A ISO/TR 7250-2:2024 Mean Elbow Height for Females and Males in the Netherlands [5]

Data pertaining to a North American population was unavailable. The Netherlands is one of the few countries within the standard containing data relating to mean elbow height and hence these values were used.

Table 12 — The Netherlands: Statistical summary

No.	ISO 7250-1:2017			Sample size <i>n</i>	Mean	SD	P1	P5	P50	P95	P99
	Sub-clause	Measurement									
		Age ^b	Male	564	38,5	13,4	18	19	38	60	64
			Female	698	38,6	12,8	18	20	38	60	64
1	6.1.1	Body mass (weight), kg	Male	595	85,6	17,3	57	62	83	117	143
			Female	679	73,9	15,8	46	53	72	103	119
2 ^a	6.1.2	Stature (body height)	Male	563	1 808,1	92,8	1 600	1 672	1 806	1 959	2 054
			Female	679	1 672,3	79,0	1 488	1 528	1 673	1 799	1 864
3 ^{ab}	6.1.3	Eye height	Male	558	1 695,9	84,4	1 488	1 565	1 695	1 842	1 916
			Female	691	1 569,1	73,2	1 397	1 447	1 568	1 689	1 749
4 ^c	6.1.4	Shoulder height	Male	542	1 476,4	81,3	1 304	1 352	1 474	1 625	1 682
			Female	587	1 361,5	70,5	1 203	1 230	1 361	1 477	1 529
5 ^{ab}	6.1.5	Elbow height	Male	562	1 122,6	66,1	965	1 018	1 121	1 239	1 296
			Female	685	1 040,7	54,4	917	949	1 038	1 126	1 181

Tinescales Causing Impatience and Impression of Latency [6]

Previous Web usage studies have found that delays greater than 10 seconds encourage users to believe that an error has occurred in the processing of their request [20]. Our finding that the threshold where QoS is judged as 'Low' is around 11 seconds is therefore consistent with previous work.

QoS stands for Quality of Service. Periods of waiting greater than 10 to 11 seconds are considered to be high levels of latency. They are instinctively associated with error, demonstrating they are perceived to be abnormal by most individuals.

[7]

- **10 seconds:** The limit for keeping users' attention focused on the dialogue. For longer delays, users will want to perform other tasks while waiting for the computer to complete the task.

The urge to complete other tasks while waiting can be considered a demonstration of impatience. This suggests periods of waiting greater than 10 seconds cause feelings of impatience in most individuals.

[8]

Brooms

My mom washes her broom with dish soap and warm water in the shower once a month. While the cleaning experts we spoke with did not recommend going to these lengths to clean your broom, they did recommend shaking out or vacuuming the bristles to get rid of any stuck-on dirt and replacing the broom when it starts to show signs of wear, usually one to two years after regular use.

[9]

According to a 2022 Statista poll, about half of the Americans surveyed (50.1%, to be exact) admitted to replacing their brooms once every two years.

Existing Hand-Held Brooms (EHHB) Considered [12-16]

Mastercraft Wide Angle Broom with Metal Handle, 48-in [12]

Specifications

Part Number	C77802
Broom Type	Angle
Handle Length (in)	53.94 in
Bristle Material	Plastic
Bristle Colour	Black
Handle Material	Wooden
Handle Colour	Brown
Broom Block Shape	Rectangle
Brush Size/Sweep Face (in)	9.45 in
Assembled Height (in)	1.38 in
Assembled Width (in)	11.81 in
Assembled Length (in)	55.12 in
Assembled Weight (lb)	1.87 lb
Handle Diameter (in)	0.94 in
Accessories List	No Accessories Included
Bristle Hardness	Soft
Wet/Dry Bristles	Wet/Dry Bristles

Amazon Basics Heavy-Duty Broom, Blue and White [13]

Product information

Technical Details

Item Weight	0.85 Kilograms
Batteries Required	No
Included components	Broom Handle
Import	Imported



Heavy Duty Broom, 55 Inch Long Handle Outdoor Large Angle Brooms [14]

Product information

Technical Details

Brand	macena
Model Number	K064-B6
Colour	Black-red
Product Dimensions	35.05 x 35.05 x 139.7 cm; 544.31 g
Material	Plastic
Special Features	Angled Bristle, Heavy Duty, Rust-Resistant
Item Weight	544 g



Libman Heavy-Duty Angle Broom [15]

Specifications

Dimensions

Assembled Depth (in inches)	2.0	Assembled Height (in inches)	52.0
Assembled Weight (in lbs)	2.0	Assembled Width (in inches)	15.0
Packaged Depth (in inches)	2	Packaged Height (in inches)	50
Packaged Weight (in lbs)	2.333	Packaged Width (in inches)	13

Details

Bristle Material	Polypropylene	Bristle Softness	Hard/Stiff
Broom Type	Angle Broom	Features	Reusable, Unscented, Grip Handle
Indoor/Outdoor	Indoor/Outdoor		

Vileda 3 Action Broom [16]

Specifications

Dimensions

Assembled Depth (in inches)	2.244	Assembled Height (in inches)	51.181
Assembled Weight (in lbs)	1.014	Assembled Width (in inches)	11.811
Packaged Depth (in inches)	2.244	Packaged Height (in inches)	51.181
Packaged Weight (in lbs)	1.014	Packaged Width (in inches)	11.811

Details

Bristle Material	Polyethylene	Bristle Softness	Medium
Broom Type	Angle Broom, Broom	Features	Reusable, Unscented
Indoor/Outdoor	Indoor/Outdoor		

Average HHB Width Calculation

$$\text{Average Broom head Width} = \frac{11.81 + 14.6 + 13.7 + 15 + 11.81}{5} = 13.38 \text{ in} \approx 33.99 \text{ cm}$$

Average HHB Weight Calculation

$$\text{Average Broom Weight} = \frac{0.848 + 0.85 + 0.544 + 0.907 + 0.4599}{5} = 0.72178 \text{ kg} \approx 0.722 \text{ kg}$$

Recommended Weight of Hand Tools [17]

What are the major ergonomic concerns of a hand tool design?

[Back to top](#)

Weight of the tool

Ideally, a worker should be able to operate a tool with one hand. The weight of the tool may depend on the use:

- 2.3 kg (5 lb) if the hand tool will be used away from the body or above shoulder height.
- 1.4 kg (3 lb) or less for tools operated with one hand
- 0.4kg (1 lb) for precision tools to allow for good control.

Justification for 75° Angle [18]

When using a leaning ladder to carry out a task:

- only carry light materials and tools – read the manufacturer’s labels on the ladder and assess the risks;
- don’t overreach – make sure your belt buckle (or navel) stays within the stiles;
- make sure the ladder is long enough or high enough for the task;
- don’t overload the ladder – consider your weight and the equipment or materials you are carrying before working at height;
- check the pictogram or label on the ladder for any advisory information;
- make sure the ladder angle is at 75° – you should use the 1-in-4 rule (ie one unit out for every four units up – see Figure 1).



Figure 1
Ladder showing the correct 1 in 4 angle

Justification for Coefficient of Static Friction [19]

Friction Coefficient Summary Table:

Material Combination	Static Friction Coefficient
soft rubber on dry concrete	0.85
hard rubber on dry concrete	0.6
soft plastic on dry concrete	0.6
hard plastic on dry concrete	0.3
soft rubber on dry wood	0.95
hard rubber on dry wood	0.7
soft plastic on dry wood	0.7
hard plastic on dry wood	0.4

4.1 & 4.2 - Aluminum as Material in Suction Cup and Tripod Designs [20]

Aluminum alloys are economical in many applications. They are used in the automotive industry, aerospace industry, in construction of machines, appliances, and structures, as cooking utensils, as

covers for housings for electronic equipment, as pressure vessels for cryogenic applications, and in innumerable other areas. Tables 6 and 7 list typical applications for some of the more commonly used wrought and cast alloys, respectively.

Table 2 Strength ranges of various wrought aluminum alloys

Aluminum Association series	Type of alloy composition	Strengthening method	Tensile strength range	
			MPa	ksi
1xxx	Al	Cold work	70–175	10–25

Aluminum has a density of only 2.7 g/cm³, approximately one-third as much as steel (7.83 g/cm³). One cubic foot of steel weighs about 490 lb; a cubic foot of aluminum, only about 170 lb. Such light weight, coupled with the high strength of some aluminum alloys (exceeding that of structural steel), permits design and construction of strong, lightweight structures that are particularly advantageous for anything that moves—space vehicles and aircraft as well as all types of land- and water-borne vehicles.

Justification for Dimensions of Tripod [21]

30. Principles of stability – State the following:

- The nearer the centre of gravity to the middle or centre of the base of support, the greater the stability.
- The wider or broader the base of support, the more stable the body.
- The lower the centre of gravity to the base of support, the greater the stability.

This indicates that the wider the caregiver's base of support and the lower their centre of gravity, the greater their ability to remain stable when handling the client. Employees must make individual adjustments based on their general conditioning, height, size, and shape.

4.3 Weighted Base Material - HDPE [28]

1. Durability and Strength

HDPE is incredibly durable, making it suitable for applications where strength and longevity are crucial. Its ability to withstand heavy loads, harsh weather conditions, and impact makes HDPE a valuable choice for construction, industrial, and outdoor uses.

2. Chemical and Corrosion Resistance

One of the standout features of HDPE is its resistance to a wide range of chemicals, including acids, bases, and organic solvents. This property makes it an ideal material for pipes, containers, and tanks that come into contact with various chemicals, as it does not degrade easily when exposed to corrosive substances.

3. Lightweight

Despite its strength, HDPE is lightweight. This characteristic makes it easy to transport, handle, and

install, reducing overall costs and improving efficiency in industries like construction, packaging, and plumbing.

4. Environmentally Friendly and Recyclable

HDPE is one of the most widely recycled plastics. It can be repurposed for a variety of products after its initial use, reducing waste and contributing to sustainability. Recycled HDPE is often used in products like piping, plastic lumber, and outdoor furniture.

5. Cost-Effective

HDPE is generally affordable, especially considering its durability and longevity. Since it requires minimal maintenance and has a long service life, HDPE is a cost-effective solution for various applications, from packaging to infrastructure projects.

6. Temperature Resistance

HDPE can withstand both high and low temperatures, making it a versatile option for outdoor applications where temperature fluctuations can be extreme. It performs well in cold weather without becoming brittle and can withstand high temperatures without warping.

7. Non-Toxic and Safe for Food Contact

HDPE is non-toxic and safe for use with food and water, making it ideal for packaging applications and water pipes. It meets FDA regulations for direct food contact, which is why it's commonly used in milk jugs, juice bottles, and food containers.

4.3 Weighted Base Attachment Weight [29]

Weight of the tool

Ideally, a worker should be able to operate a tool with one hand. The weight of the tool may depend on the use:

2.3 kg (5 lb) if the hand tool will be used away from the body or above shoulder height.

1.4 kg (3 lb) or less for tools operated with one hand

0.4kg (1 lb) for precision tools to allow for good control.

4.3 Weighted Base Inner Material [30]

Silicone rubber is a synthetic rubber having stable elasticity and weather resistance in a wide temperature range. It has relatively high elongation performance and is widely used as a shock absorber and sealing material, but its tensile strength may be slightly lower than that of other rubbers. In addition, due to the slightly sticky surface, friction tends to occur easily in a dry state. Because it is resistant to temperature changes and retains the flexibility even under high temperatures, it is used in a wide range of purposes

from the medical and food fields to automotive parts and electronic equipment. It also has excellent weather resistance and electrical insulation properties, and the moderate surface friction prevents slippage. However, when requiring sliding, it is necessary to have a special surface treatment and material selection. At the same time, the sealing surface has a feature of sufficient airtightness, so it is expected to be used in a variety of fields.

4.3 Weighted Base Overturning Factor of Safety [31]

The overturning safety factor (OSF) is the sum of resisting moments divided by the sum of overturning moments. Most codes require that this factor be greater than 1.5. Overturning safety factor calculations are based on the service load combinations only and are calculated in both the X and Z directions.

4.4 Magnetic Stand Weight [32]

Brand	SavorSpace
Colour	Black and Orange
Parcel Dimensions	39.4 x 29.9 x 10.3 cm; 1.04 kg
Material	Metal, Plastic
Special Features	Compact Storage: Broom clips neatly onto the dustpan handle for space-saving storage., Durable Materials: Reinforced with high-quality plastic and stainless-steel parts to withstand frequent use., Ergonomic Long Handle: Prevents back strain with a convenient, upright design for comfortable cleaning., Heavy-Duty Broom Bristles: Captures everything from fine dust to large debris without leaving residue behind., Snap-Lock Dust Pan: Secures contents to avoid accidental spills during disposal.
Item Weight	1.04 kg

4.4 Magnetic Stand Material - HDPE [28]

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2. Chemical and Corrosion Resistance

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HDPE can withstand both high and low temperatures, making it a versatile option for outdoor applications where temperature fluctuations can be extreme. It performs well in cold weather without becoming brittle and can withstand high temperatures without warping.

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HDPE is non-toxic and safe for use with food and water, making it ideal for packaging applications and water pipes. It meets FDA regulations for direct food contact, which is why it's commonly used in milk jugs, juice bottles, and food containers.

4.4 Neodymium Magnets Magnetic Stand [33]

Neodymium magnets are the most powerful permanent magnets that exist today. They are made of a combination of neodymium, iron, and boron, and are commonly referred to as NdFeB magnets. These magnets exhibit incredible strength, **with some grades capable of holding over 1,000 times their own weight.**

The strength of neodymium magnets is measured in units of magnetic energy product, or maximum energy product, which is calculated by multiplying the remanence (the magnetic induction remaining in a material after it has been magnetized) and the coercivity (the ability of a magnetic material to resist being

demagnetized). The higher the maximum energy product, the stronger the magnet.